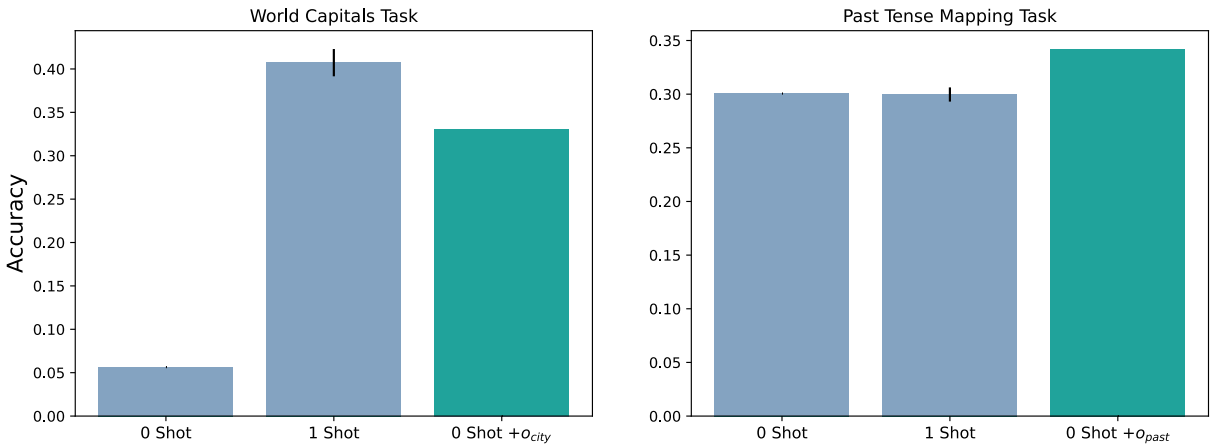

Figure 14: By replacing FFN networks with the corresponding $\vec{o}$ vectors, we show that we can improve zero-shot performance by taking advantage of the model going through argument formation in the zero-shot setting.

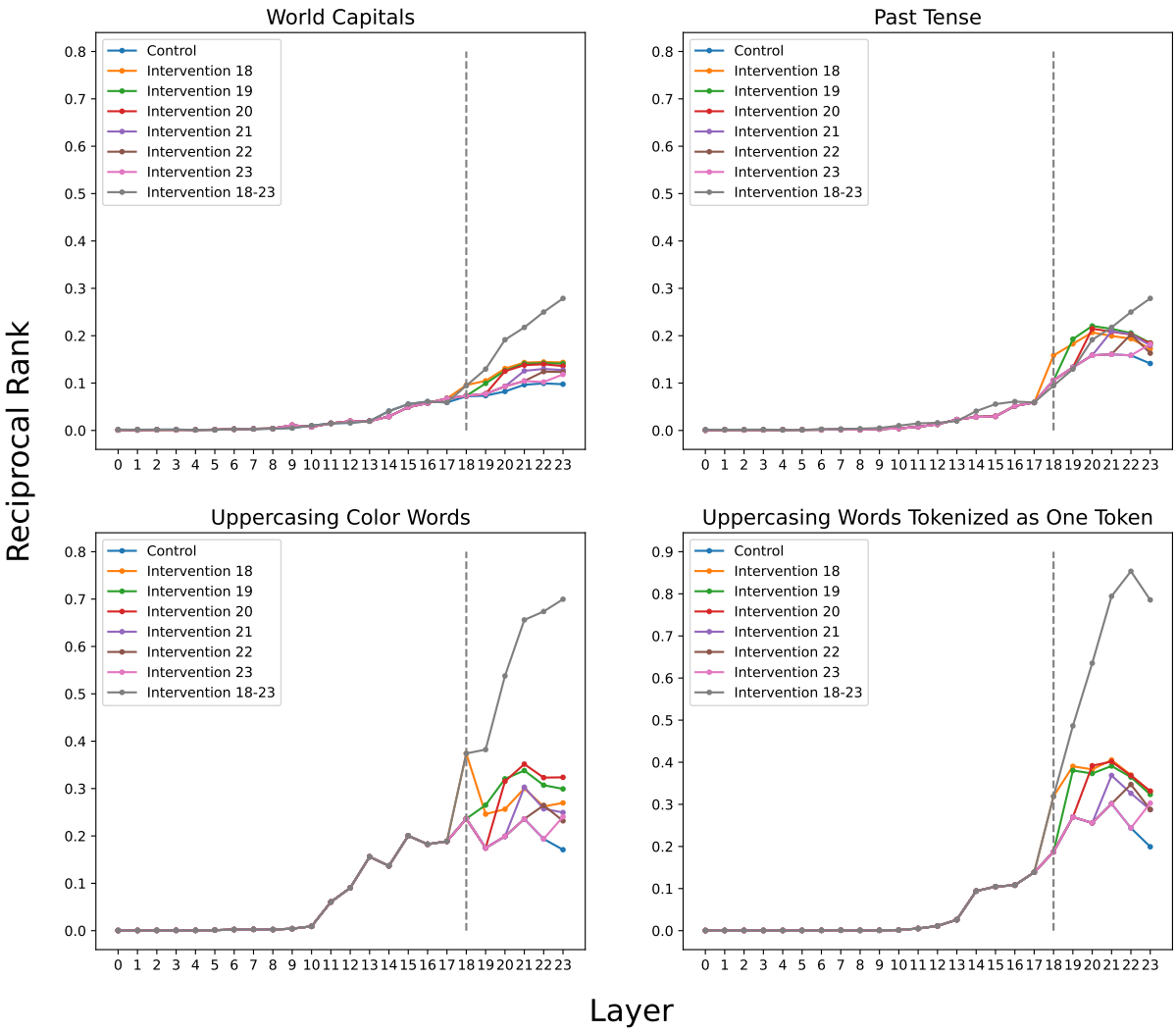

Figure 15: Replacing any individual FFN update is worse than replacing all of them. This supports the idea that networks made gradual updates to their representations, and that the $\vec{o}$ vectors we extract behave this way as well: multiple similar updates are made k layers in a row. Interestingly, the average boost to the reciprocal rank is about the same regardless of which single layer we apply the update at, suggesting that this range of FFNs are operating in same space.

that the embeddings would encode a linear relationship between these two, since “Pur" is a subtoken in many other words. We explore this specific phenomenon on the random tokens task from Section 4 with the $o_{\vec{upper}}$ intervention. We take 100 single token words that capitalize to a single token, and 100 others that capitalize to words that break down into multiple tokens. Our results can be seen in Figure 16. We find that tokens that get broken up into multiple tokens are less probable than for tokens that capitalize to single token forms.

G Additional Tasks: One-to-One, Many-to-One, and Many-to-Many Relations

In the main paper, we show study three one-to-one relations that exhibit the argument/output pattern, but it remains unclear how well this generalizes to other relations. Using six additional tasks, three many-to-X and three new one-to-one, we provide evidence that suggests that the observed mechanism is specific to one-to-one relations, and does not work when multiple inputs map to one output. This suggests that the model is sensitive to this distinction of relations during pretraining, and the vector arithmetic mechanism structure we observe only presents for the most explicit relations. In Table 2, we give examples of the six new tasks, following the same prompt format as the one used in the main paper. In Table 3, we break down the relation type of each task and provide the GPT2-Medium accuracy for each one. Figure 17 shows the early decoding patterns for the argument and answer tokens. While the three one-to-one tasks exhibit the initial promotion of the argument token, followed by the answer token on average, the argument token does not become highly promoted on any of the *non* one-to-one relations.

H Compute

All models were run on NVidia RTX 3090s; Bloom was run locally on 3090s in float16 with CPU offloading.

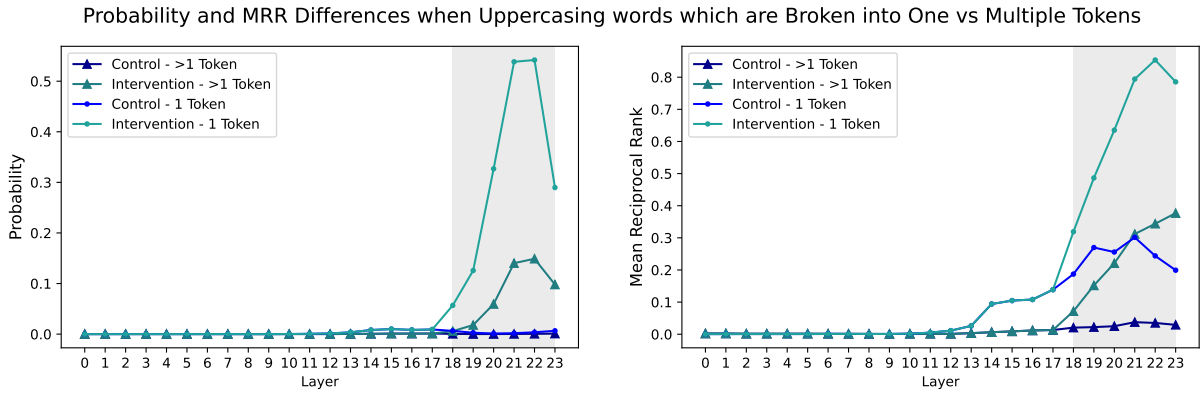

Figure 16: When the uppercase version of a word gets broken down into multiple subtokens, mapping to that token becomes much less probable and is generally harder of an association for the model to make.

Task	Example
Animal Hypernyms	...Q: The anaconda is a kind of what?\nA: (snake/reptile/boa/...)
Name to Nationality	...Q: What is the nationality of Balzac?\nA: (French)
Country to Language	...Q: What is the official language of Argentina?\nA: (Spanish)
Adj. to un-Adj.	...Q: What is the opposite of able?\nA: (unable)
3rd Person Verbs	...Q: What is the third person singular of become?\nA: (becomes)
Noun Plurals	...Q: What is the plural of album?\nA: albums

Table 2: Examples from three non-injective and one injective relation. A given animal (anaconda) is a type of snake and reptile, and other snakes/reptiles also exist (many-to-many). Balzac is only French and other people map to French (many-to-one), etc.

Task	Accuracy (%)	Task Type
Animal Hypernyms	30.4±1.7	Many-to-Many
Name to Nationality	73.2±2.0	Many-to-One
Country to Language	71.2±2.4	Many-to-Many
Adj. to un-Adj.	12.0±1.1	One-to-One
3rd Person Verbs	22.4±0.7	One-to-One
Noun Plurals	51.6±1.7	One-to-One

Table 3: One-shot accuracies for each task across 5 random seeds for GPT2-Medium.

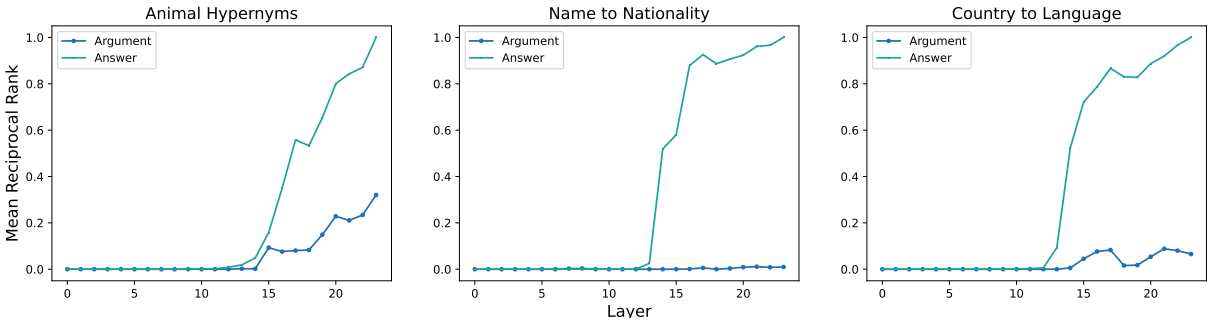

Figure 17: Non-injective tasks show no evidence of argument-function processing on average. In sharp contrast to the past tense, colored objects, capital cities, and un-adj. tasks where this is observed, here, the argument token experiences virtually no spike in reciprocal rank in the intermediate layers.